Matt Amor
ECON 6376 – Fall 2026
Midterm

Section 1 - Introduction

Is there a relationship between the distribution of home prices and the distribution of income? Income certainly relates to home prices. Someone with a job offer in an expensive city will negotiate for higher salary, and a home seller will price a property as high as the market of local buyers can afford. But how does income *inequality* relate to the *distribution* of home prices? When a country's income grows more concentrated, do the expensive homes get more expensive, and does that drive up the cost of all homes?

Imagine a small group of people with income high enough to purchase the most expensive homes, and they compete over a stock of housing that is too small (as has been the case in most of the housing market). In this case sellers have market power and raise prices to match demand, much like the broader housing market. This would likely result in a steady distribution of housing prices.

In contrast, the richest buyers could look at a broader market of homes than is accessible to most people. This raises their effective market power, driving down prices at the high end. An indicator marking the difference between this example and the previous one is that the average price of a home here is lower, while the median home sold at the same price. This means the spread between the average and median home prices decreased compared to a market that is more stratified.

A complication with the stratified scenario is that rich people may be less inclined to bargain hunt and more inclined to pay higher markups on luxury, both of which raise prices of high-end real estate relative to the rest of the market. This increases the spread between average and median home prices.

The necessary data to see this effect over time is average home sale prices, median sale prices, the share of income of the top 10% of income earners, and active listing counts. The average and median home sale prices need to be examined since the question specifically relies on their relationship. The sales prices (as opposed to listing prices) are required since listing prices are *estimates* of what a market can accept. If there is a relationship between the buying habits of the wealthy and the overall distribution of the housing market, it likely manifests in confirmed market activity. Inequality is measured by the share of income of the top 10% of income earners because this group is likely to interact with the highest-priced outliers in the housing distribution, pulling up the average. Active listing counts help indicate if a market is largely a "buyer's market" in which there is downward pressure on prices, or a "seller's market" in which the pressure on prices is upward.

Section 2 – Literature Review

The relationship between income inequality and housing prices has some support in the literature. Rodda (1994) outlines a framework for linking the income distribution to rents of the poor. On page 32, Rodda proposes a possible model for understanding how a housing market is differentiated between rich and poor consumers. This leads to a situation where rich consumers upgrade as their incomes increase, while the poorer consumers move into the high-quality units left behind. This market-wide upward mobility only occurs as long as the high-income consumers move. Rodda finds a significant positive relationship between high incomes and rent increase for low-cost units.

Matlack and Vigdor (2008) expand on that analysis and argue that how much impact inequality has on housing costs varies across markets. They examined households headed by people without a high school diploma from 1970 -2000, holding constant year and location fixed effects. Where income inequality is high *and* vacancy rates are low, they find that poorer people do pay higher rent compared with low vacancy rate markets with lower inequality. "Slack in the housing market," they say, "appears to mitigate any negative impact of income inequality increases on housing consumption of the poor." If price pressure from growing inequality to the rent of poor people is dependent on vacancy rates, this suggests the mechanism uses the distribution of housing prices to transmit that pressure. This transmission is what this paper seeks to illuminate.

One motivation for the question of this paper is to identify if there is transmission of price pressure from the top of the home price distribution down. Existing literature has described the link between income inequality and rent paid by the poorest people, but the mechanism through which this can happen remains nebulous. Does this happen because wealthy people drive up the costs of the most expensive homes, and home prices induce higher rent? Do they own property then extract rent directly from low-income households? Or are more things happening at once?

Section 3 – Data

The data selected for the series comes from the US Census Bureau, the US Department of Housing and Urban Development, and Realtime Inequality. The series for Average Sales Price of Houses Sold for the United States and Median Sales Price of New Houses Sold for the United States came from the Census and HUD. While the average and median values come from the same table directly from the Census, the identical series data available from the Federal Reserve Bank of St. Louis (ASPUS and MSPUS) require less cleaning. The spread between average and median home prices is derived from the previous two series. Specifically, this is the difference between the average and median home prices divided by the median home price for a given quarter. Since this measure is always a proportion of the given quarter's median price, this removes the effect of inflation from the measure. The measure of inequality (share of income for the top 10% of income earners) came from Realtime Inequality, an organization providing high frequency data on income inequality. Quarterly housing inventory is produced by the Census with its monthly Housing Vacancy Survey.

Table 1 (see appendix) shows the data summaries of the four raw series as well as the spread of average versus median home prices, with links to their sources. The inequality and inventory series each have one extra observation (starting their series one quarter ahead) so that, when they are differenced, they still include the same number of observations. Table 1 shows where the time series come from, while *data.csv* displays the final transposed time series values.

Average and Median Home Prices

Both the average and median home price series come from the Survey of Construction (SOC) released by the US Census Bureau and partially funded by the US Department of Housing and Urban Development. The two series come from the same table called the "Median and Average Sales Price of New Homes Sold by Region" available on the Historical Time Series section of the Census' New Home Sales website.

The data on home sales is available as a time series beginning in 1963. The survey is conducted monthly using data from owners/builders of residential housing construction. The inclusion of worksites is determined by randomly selecting permits from 900 permit offices. In the case of the 70 areas examined by the Census Bureau that do not require building permits, surveyors walk along roads in areas until they get to a lot that appears to be under construction. Every new home found in the canvassed area will be selected for the survey.

Each month's release is preliminary followed by three sets of revisions with the subsequent months' releases. After the third revision, estimates are no longer revised, but

the seasonal adjustments are updated annually in the April release and can affect the previous 63 months.

The estimates recorded from the survey are adjusted by the total number of authorized housing units obtained by the Building Permits Survey to extrapolate estimates for the whole country. The data are then seasonally adjusted. Quarterly totals are included with the release of the final month of the quarter.

As indicated by the name of the release, the data recorded are prices only for newly constructed homes owned by private entities in the United States. Since existing homes are excluded, this raises a major threat to the soundness of this analysis which is bias in the data. It may be the case that wealthier buyers are more likely to have homes built to their specifications instead of buying whatever is available on the existing homes market. Since the measure of inequality is the relative richness of the highest earners, the skew of the new home price distribution might increase from selection bias rather than overall housing market forces.

For the purposes of this report, the average and median prices are converted to 2026 dollars using the CPI-U. More detailed information about the Methodology of the Survey of Construction is available from the Census.

Skew of Housing Distribution

To estimate the skew of the housing price distribution, a measure of spread is derived from the average and median home prices. This is a simple formula:

$$Skew = \frac{(Average\ Home\ Price - Median\ Home\ Price)}{Median\ Home\ Price} * 100$$

This is reported as a percentage of the median home price for each quarter. Since outliers influence average values more than the median, a wider gap between the average and median, while holding the median constant, implies that outliers are more influential. In the context of housing, the influential outliers are high-priced homes sold to wealthy (generally high income) households. What a large gap could imply is that there is segmentation in the upper end of the housing market. If the gap is narrower, this could mean a smoother gradient between expensive homes, mansions, and estates.

Income Share of Top 10% of Income-Earners

The measure of income distribution used is the share of the top 10% of income-earners, retrieved from the organization, Realtime Inequality. The Census reports detailed information about the distribution of income in the United States but only does so annually. Realtime Inequality derives certain measures of income inequality from a group of other statistics that are reported monthly.

Available metrics include, Income of Middle 40%, Share of Income of Bottom 50%, Disposable Income for Top 0.001%, and so on. This dataset focuses largely on the highest brackets of income-earners, so the reporting of lower percentile groups is not exhaustive (70th percentile – 90th percentile is absent) or mutually exclusive (Bottom 50% includes half of the Middle 40%). This makes the Top 10% the ideal metric for this analysis, since it is large enough to have an impact (23.5 million in 2026), but specific enough only include individuals with enough income to affect the home price distribution.

The monthly data span from January 1976 through June 2026 and quarter values were obtained by averaging across each quarter. There are three measured populations which are adult individuals (ages 20 and up), working age adults (ages 20 through 64), and households (combining married people and dependents together as one unit). The unit selected for this analysis is households, since they tend to purchase houses together.

There are four measures of income. Factor income includes all labor and capital income. According to Realtime Inequality, this is the "most comprehensive and harmonized notion of income," and, since many wealthier individuals can leverage their capital wealth to purchase homes, this is likely to be the measure most influential for the relationship to housing distribution. There is also "pretax income" subtracting pension, disability, and unemployment insurance contributions (and adding any benefits accrued by these), "disposable income" which nets out taxes/government transfers, and "posttax income" which adds collective public expenditures and removes the deficit.

The data are estimated from monthly household and employment surveys, quarterly surveys and administrative records, the Quarterly Census of Employment and Wages, and "synthetic microfiles." Observations are simulated adults (with income, wealth and the ingredients that comprise both) based on the work of Piketty, Saez and Zucman (2018), combined with the Current Population Survey and the Survey of Consumer Finance to add in demographic information.

It is worth emphasizing that this entire series (reaching back to 1976) is based on synthetic data to achieve the higher frequency values. From page 4 of the working paper outlining their methodology, the authors state, "Comparing predicted to observed income changes, we find that we correctly anticipate whether income is growing or falling 89% of the time for the top 1%, 93% for the next 9%, and 82% for the next 40% and for the bottom 50%." This bolsters the selection of the top 10% as the metric to use here. The methodology was developed by Thomas Blanchet, Emmanuel Saez, and Gabriel Zucman and is detailed in a working paper published by NBER.

Housing Inventory

The housing inventory data come from table 8 of the Housing Vacancy Survey (HVS), which is a monthly supplement to the Current Population Survey. It includes values from 1965 through the Q2 of 2026. The current sample includes 72,000 housing units and surveys are conducted once a month. Households are surveyed every month for four months, then not at all for eight months, then surveyed again for another four months, before being permanently removed from sample.

Since 1981 (the beginning of this paper's sample), the CPS has revised certain attributes of the survey including how they estimate sample ratios. Pages 32 through 38 of the 2019 TP77 report on the CPS design and methodology document the notable changes to how the survey has been conducted throughout its history. Relevant points in that history include April 1994, when the monthly sample fell from 72,000 to 68,000 before getting cut again to 59,500 in January of 1996. The 1996 reduction was the result of relaxed reliability requirements which stemmed from cut in funding. In July of 2001, the monthly sample size was increased to 72,000 units. Then in January 2003, the CPS incorporated new standards derived from the 2000 Standard Occupational Classification (SOC) and the 2002 North American Industry Classification System (NAICS). This was a major change in the CPS but appears not to have impacted the national housing vacancy data. Then in 2014, a sample redesign led to another increase to 74,000 households, but by 2019 had been reduced back to 72,000. The shifts in sample size of the CPS could impact the accuracy of the estimates from the mid 1990s to 2001.

The data taken from the HVS table 8 is the "For sale only" subsection of the national total of vacant units for each quarter, and the sample included in this report is from Q4 of 1980 through Q4 of 2019. The sample starts one period early to allow for a differenced series to align with the other data. This should be likely to scale with population, but the data show inventory has been steadily declining since the 2008 financial crisis.

Housing inventory is included in this analysis because home prices are likely to have an inverse relationship to the number of homes that are for sale. More houses on the market allow buyers to choose less expensive and better suited properties, which puts downward pressure on prices. By including the national inventory of homes for sale, we may be able to control for other factors impacting the distribution of prices.

Section 4 – Stationarity

For time series analysis, the most important data processing step is determining whether the series is *stationary* or not. A stationary series enables an econometrician to approximate statistics from its realizations, but they have a few requirements. The series in question needs to have a constant mean, constant variance, and stable autocovariance to be considered *weakly stationary*. Formally, these three conditions are:

$$E[y_t] = \mu \text{ for all } t$$

$$Var(y_t) = \sigma^2 \text{ for all } t$$

$$Cov(y_t, y_{t-k}) = \gamma_k \text{ depending on gap } k$$

A series will also meet the definition of *strict stationarity* if it meets all the definitions of weak stationarity and if every finite dimensional joint distribution is unchanged by a common time shift.

$$F(y_{t1}, \dots, y_{tn}) = F(y_{t1+h}, \dots, y_{tn+h}) \; \forall h$$

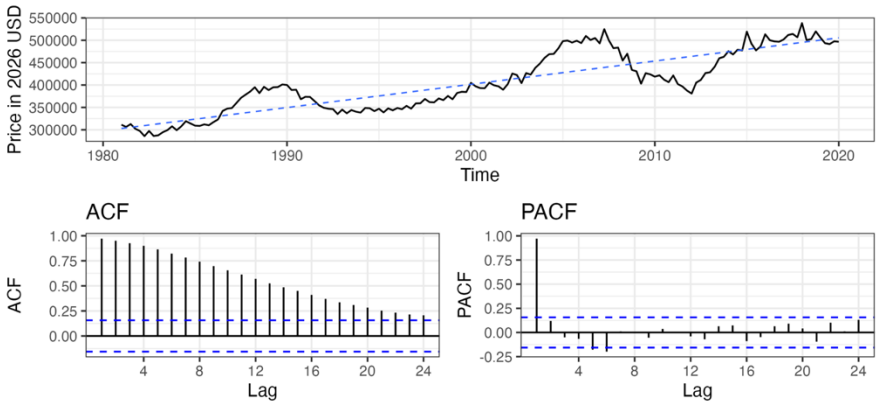
What this means is that any sample taken from a strongly stationary series should look indistinguishable from any other sample. The strict stationarity requirement is both too restrictive and unhelpful when analyzing economic data, so any further mention of stationarity here refers only to weak stationarity.

If a series is not stationary traditional statistical methods can draw conclusions that look compelling, but only because the series failed to meet the assumptions of those methods. This can make significant relationships appear where they do not exist. So before drawing conclusions, stationarity has to be determined, and necessary transformations have to be performed on non-stationary series.

There are three major steps to test if a series is stationary. The first is to look at the raw series, next is to examine the autocorrelation functions, and finally perform formal tests. Figure 1 shows the raw time series data of the average inflation-adjusted home prices. This is not a stationary process as it appears.

A non-stationary process is often called a random walk and has a persistent memory. The memory of a non-stationary process is referred to as a unit root, which has distinct characteristics in the plots from Figure 1. The series plot (the wide top panel) of a process with a unit root will wander away from a long run mean and fail to return. The autocorrelation function (ACF) pictured in the bottom left panel will decay very gradually and linearly, and the partial autocorrelation function (PACF, bottom right) will show one very high lag before dropping off completely.

FIGURE 1 - Average Home Prices 1981 - 2019



At first glance, Figure 1 looks like it matches all the visual descriptors of a unit-root process, but the top panel signals a possible fix. The series does not have a *constant* mean, but it might have *trending* one as indicated by the dashed blue line. If the series were transformed to correct for the trend, it could look stationary. Figure 2 shows the detrended series.

The detrended series looks like the graph in Figure 1 if it were turned clockwise a bit. It still appears to have a very long memory, which is evident from the deviation from the mean in the early 1990s, only returning in 2000. The ACF is still gradually decaying at a fairly linear rate, but the PACF shows two significant initial lags, neither of which are as close to 1 as the earlier PACF. The PACF also shows several significant later lags. Overall, this is visually inconclusive, so it helps to look at formal test results.

FIGURE 2 - Detreneded Avg. Home Prices

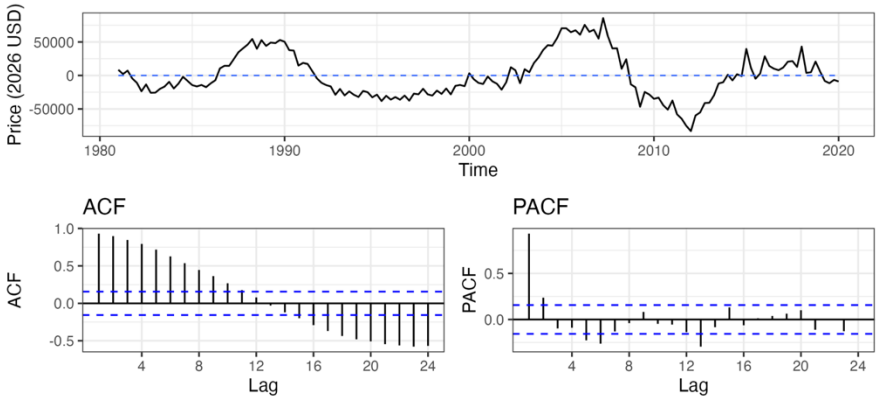


Table 2 shows the results of two tests for each of the three versions of this series: the untransformed average home prices, the detrended average prices, and the first-differenced average prices. The null hypothesis for the Augmented Dickey-Fuller test is

that the series is generated by a unit-root process, while the null for the KPSS test is that the series is stationary. Since they test for opposite things and we want a stationary series, the ideal situation is a rejection of the null of the ADF test, and failure to reject the null of the KPSS.

TABLE 2 - Unit-root and stationarity tests

Test	*Null*	*Statistic*	*Critical values*			*Decision at 5%*
			1%	*5%*	*10%*	
Average Home Prices						
ADF	Unit root	−0.936	−3.460	−2.880	−2.570	Fail to reject
KPSS	Stationary	2.500	0.739	0.463	0.347	Reject null
Detrended Avg. Home Prices						
ADF	Unit root	−1.789	−3.460	−2.880	−2.570	Fail to reject
KPSS	Stationary	0.107	0.739	0.463	0.347	Fail to reject
Differenced Avg. Home Prices						
ADF	Unit root	−9.073	−3.460	−2.880	−2.570	Reject null
KPSS	Stationary	0.058	0.739	0.463	0.347	Fail to reject

In the untransformed series, the results of the ADF and KPSS were “fail to reject” and “reject” respectively. Combined with the visual evidence, it could be tempting to write the series off as non-stationary, but if detrending the series brings it into stationarity, it would be a “trend-stationary” series. Looking down at the results for the detrended series, there is a conflict between the tests. The ADF is still failing to reject the unit root at any level, but the KPSS fails to reject stationarity at any level. Given the very long memory evident in the series, detrending is unlikely to change the result for the ADF.

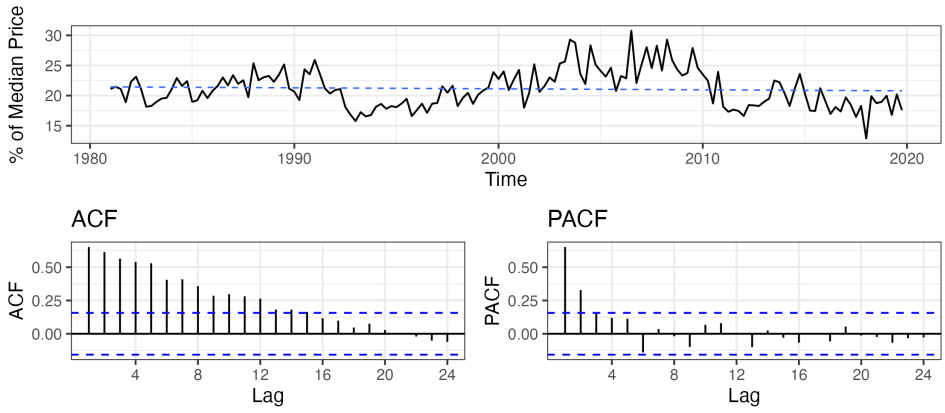
This opens up the next possible transformation, which is differencing. When a series is first-differenced (the process can be performed as many times as necessary but generally only done once) it replaces the observed y_t with the difference between y_t and y_{t-1}. This removes much of the wandering behavior of the series. After performing the first differencing on the average home price series, both tests give the ideal results, but Figure 3 (see appendix) shows the consequences of over-differencing. The ACF and PACF become muddled and lose much of their explanatory power. This will make modeling much more difficult. Since the detrended data appear to follow long steady

cycles, and we have the support of one of the stationarity tests, the average home prices series is judged to be trend stationary and the detrended series will be used in future phases of this analysis.

Moving on the to the median home prices, Figure 4 (see appendix) shows the untransformed series. The first thing to notice is that it looks very similar to the average home prices for the same years. This is not surprising since they track the same market and rely on the same responses from surveys. The series are similar enough that the analyses of their stationarity follow similar trajectories. The only slight difference is that the detrended median price series performed better on the ADF test, but it still (albeit slightly) resulted in a failure to reject the null at 5%. For the same reasons as with the average series, the median home prices will be taken as trend-stationary, and their detrended versions will proceed to the next phase.

The series derived from the average and median home prices faced different challenges than the ones faced by its constituent parts. Figure 7 shows the untransformed series. There is no obvious trend, but it does have a non-zero mean, and close inspection of the blue dashed line reveals that there is a very slight downward slope. The values appear to wander from the mean and return, but there is more volatility quarter-to-quarter. The ACF decays gradually, but not evenly or linearly and the PACF decays into insignificance after the third lag. This could be stationary without transformation, and Table 4 (see appendix) shows the test results.

FIGURE 7 - Skew of Home Price Distribution 1981 - 2019

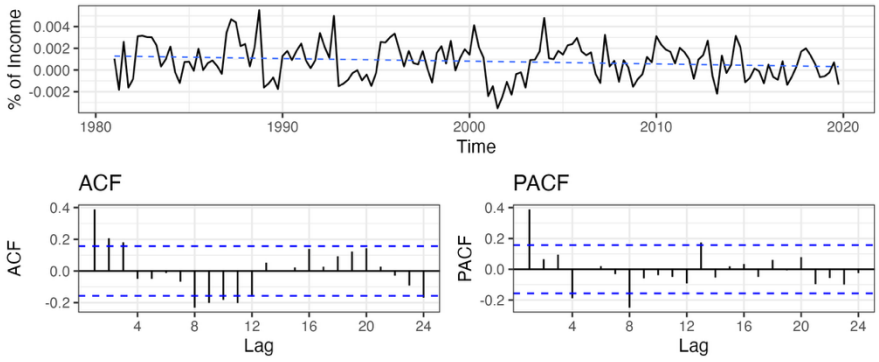


The untransformed series passes both tests, solidifying the case for stationarity. One thing to note about the ADF test on the raw series is that it was specified as the "drift" version of the test, since it had a non-zero mean. It is the zero mean of the detrended data that gives it an advantage over the untransformed data. Detrending does not remove any explanatory power like over-differencing, and later processes will have to consider the drift, so removing that now mitigates the possibility of confusion later. Figure 8 (available see appendix) shows how similar the detrended series appears next to the raw data with the key difference being the zero-mean of the detrended series.

The next series to examine is the inequality data from Realtime Inequality. Immediately evident in the plot of the raw data is a trend, so detrending is a reasonable first step. Figure 10 (see appendix) shows the detrended data.

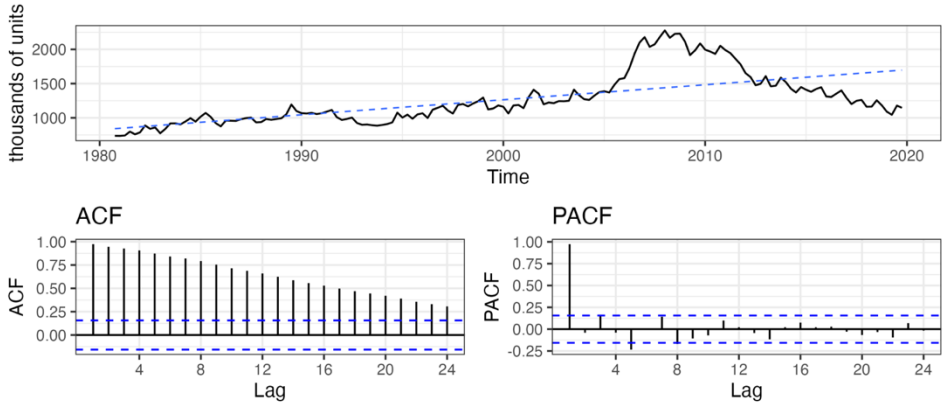
The ACF decays at a fairly linear pace, and the first lag of the PACF is nearly 1. More worrying is the slight inverted parabola of the series itself. It does not look like a series that reliably returns to a mean, but this could be an artifact of the sample window. In fact, Table 4 (see appendix) bolsters this story by showing the test results including the differenced data. Despite the apparent issues of wandering away from the mean, the detrended data passes both tests of stationarity, but it is worth noting how close the statistics are to their respective critical values. The stats for the detrended series barely fall out of the 5% threshold in the ADF test and barely fall within the 5% threshold of the KPSS. These are small enough wins that the visual inspection should carry more weight. The key fact leading to a judgement of "not stationary" is that the series displays sharp jumps followed by long periods of calm (non-constant variance) and appears ambivalent to its long-run mean (non-constant mean). Even the detrended series has an ACF with a linear decay and one very high PACF lag. To err on the side of caution, the analysis will proceed with the differenced inequality data pictured in Figure 11.

FIGURE 11 - First Differencing of Inequality



The final series to examine is the housing inventory data from the Census. Figure 12 shows the untransformed data, and it reveals two issues. The first is the familiar problem of a trend, but the second is a large increase in the early 2000s, followed by a steady decline. This is the housing bubble and subsequent bust that was one of the causes of the 2008 financial crisis. The shape of the data does not lend itself easily to a simple trend, so differencing is almost certainly the better transformation.

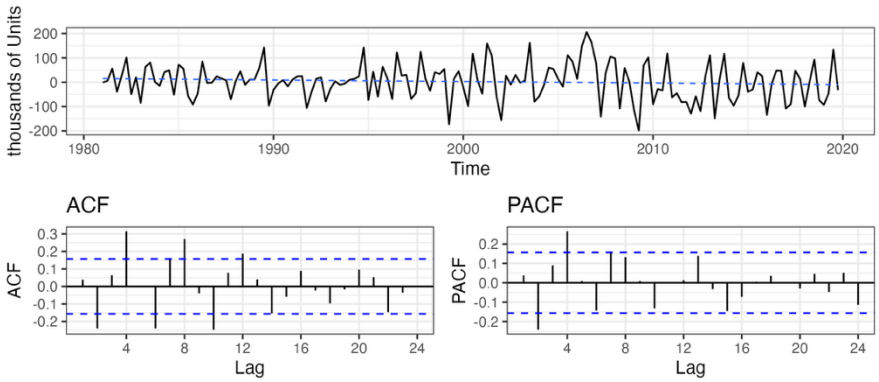
FIGURE 12 - Housing Inventory 1981 - 2019



The differenced series, shown in Figure 13, is much better behaved, and the ACF and PACF retain plenty of information about the series. They both oscillate but decay in a stable way. The test results in Table 6 (see appendix) support the decision by including the raw, detrended series, and differenced series. The raw data unsurprisingly fails both tests of stationarity, while the detrended data only fails on the ADF test. The series may

have been stationary until the early 2000s, but the massive relative spike followed by a clear consistent decline breeds distrust in the case of a comprehensively stationary process. Given the mixed test results, the visual inspection carries more weight, so this judged to be non-stationary.

FIGURE 13 - Differenced Housing Inventory

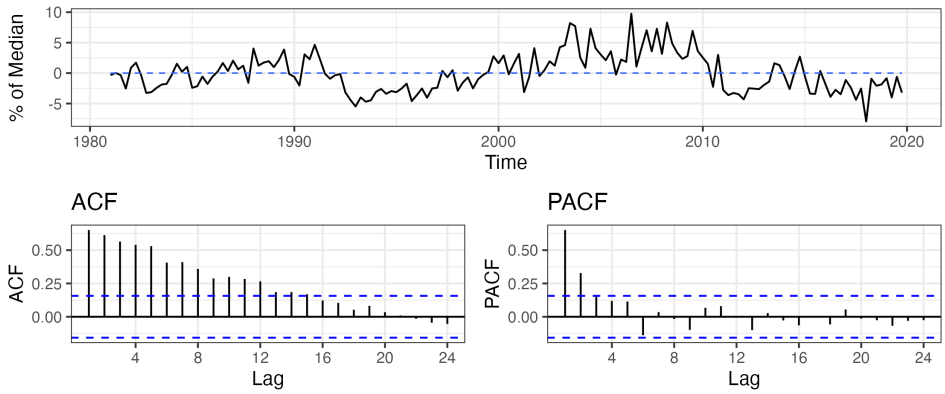


Section 5 – Model Selection

The series the rest of the report will use is the spread between median and average home prices as a percentage of the median home price. This detrended series is stationary (section 4 concluded the raw series is trend-stationary) and runs from January 1981 through December 2019. Since these are quarterly records, the total number of observations, T, equals 156. The first 132 observations (1981 - 2013) will be used to train the model, which is what this section tackles. The subsequent 12 observations (Q1-2014 through Q4-2016) will be used to weight the model, and the final 12 (Q1-2017 through Q4-2019) will eventually be used to test it.

Figure 8 (reprinted in this section) shows the full stationary data and accompanying ACF and PACF plots. The clearest options for model candidates here are AR(2) or AR(3), since the ACF decays gradually and the PACF has 2 lags (possibly a 3rd) that reaches significance in the PACF plot. A case could be made for an ARMA model, since the PACF could be described as "gradually decaying," a feature of ARMA processes. To open the analysis to the possibility that an ARMA model might fit the data better, an ARMA(1,1) and ARMA(2,1) are included as candidates.

FIGURE 8 - Detrended Skew of Home Prices 1981-2019



The end goal with this exercise is to use the model to make predictions; not describe the underlying model. Defining a model's success requires the selection of a criterion on which the candidate models will be judged. Since the primary goal is forecasting, the information criterion that rewards models best at prediction will be used. That is where the Akaike Information Criterion (AIC) excels, but it is worth looking at alternatives to bolster the case for AIC. With a small sample size, a modified AIC called AICc penalizes additional parameters harsher, but with a maximum of 4 parameters in the candidate set

and 132 observations in the training data, the difference between the two measures should be negligible. The primary alternative to AIC is the Bayesian Information Criterion (BIC), which punishes additional parameters more harshly than AIC and tends to perform better at identification of true, parsimonious models (with a high enough number of observations).

Since this analysis is less concerned with underlying truth and more concerned with accurate forecasts, the best candidate in the set will have the lowest AIC as made clear by the formal definition of the AIC.

$$\text{AIC} = -2logL + 2k$$

The key part that differentiates the AIC from alternatives is the "2k" term. This is called the penalty, and it operates as a linear function of the number of parameters. The more parameters present in the model, the higher the AIC and worse the model looks.

Table 7 reports 5 candidate models and their respective information criterion values. AICc and BIC are included as comparison points. An AR(1) [shown here as ARIMA(1,0,0)] is included, even though it was unlikely to be a good fitting model, to show the trend of AIC as more parameters are added. For each of the criteria, AR(1) displays a clear step up in value (lower values are better), illustrating how this model is unlikely to fit well. The most obvious candidate model from the ACF and PACF plots appeared to be AR(2) (with AR(3) close behind), since there was gradual decay in the ACF and 2 or 3 significant lags. Instead, models that included an MA term proved to be a better fit. The best fitting model of those tested was the ARMA(1,1) [ARIMA(1,0,1)]. It and the next two strongest candidates, AR(3) and ARMA(2,1), will be carried into the forecasting section. There is no middle *d* term for any of the candidates since the detrended data were shown to be stationary already.

TABLE 7 - Comparison of Home Price Skew Models

Model	k	AIC	AICc	BIC
ARIMA(1,0,0)	2	601.8643	601.9573	607.6299
ARIMA(2,0,0)	3	588.9016	589.0891	597.5500
ARIMA(3,0,0)	4	587.8865	588.2014	599.4177
ARIMA(1,0,1)	3	585.4556	585.6431	594.1040
ARIMA(2,0,1)	4	587.2288	587.5438	598.7600

One thing to note about information criteria in this table is that all three agree on the strongest model, but BIC ranks the two models with 4 parameters (our 2nd and 3rd place models) lower than AR(2) (4th place). This is because BIC penalizes extra parameters more aggressively than AIC and AICc. This could have been predicted by looking at the PACF plot, showing the third lag on the cusp of significance. The criterion that holds marginal parameters to a higher standard was always likely to rank AR(3) with a marginally significant 3rd lag lower than AR(2).

As a side note, the hypothesis of stationarity is rejected at the 5% level on the training set using the KPSS test for stationarity, even though the full series passed this test. What this means is that auto.arima() returns an ARIMA(0,1,1) as the best fitting model. Because the process used the KPSS test to mark the series as non-stationary, it immediately narrowed the candidates to ones that have a differencing term and found that AIC is minimized at about 584. When setting the parameters to stationary = TRUE and (just to cover all bases) stepwise = FALSE, it confirms ARIMA(1,0,1) has the lowest AIC in its search. Using the trace = TRUE also shows that Table 7 captures some of the best performing models.

Section 6 – Serial Correlation

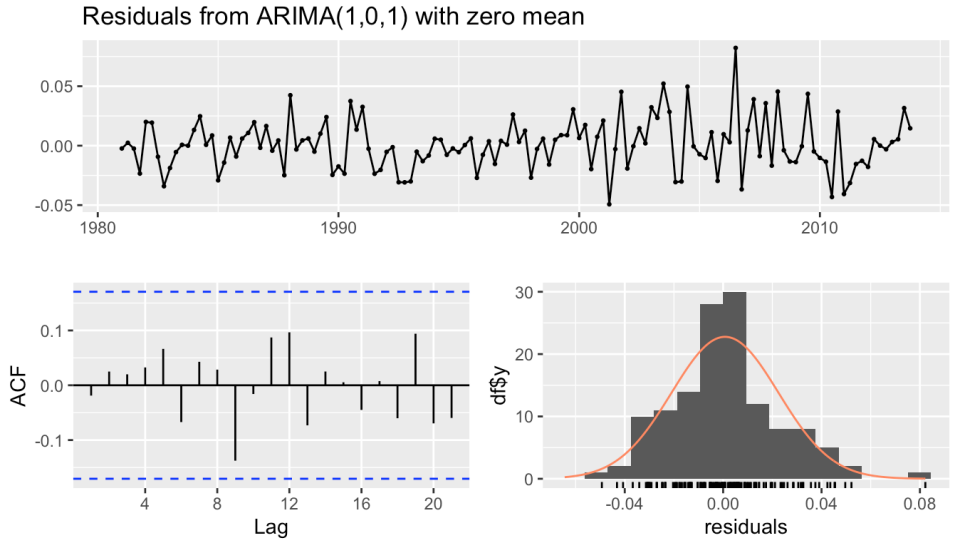
Serial correlation is a property of a model that does not fit the data properly. It happens when the residuals of the model, *e*, show significant autocorrelation. Ideally, the residuals should meet the following criteria:

$$E[e_t] = 0$$

$$Var(e_t) = \sigma^2$$

$$Cov(e_t, e_{t-k}) = 0 \text{ for all } k \neq 0$$

If the residuals of a model fail to meet these criteria it means the model failed to capture something significant in the data, often a seasonal effect. This is tested by looking at the ACF of the residuals and checking for significant lags. Incidentally, when doing this for all three of the top-ranking models, the plots look exactly like they should and very similar to one another (see appendix). The residual plot of the modeled series (the top panel) looks like white noise scattered around a zero mean, the ACF has no lags that pass the threshold of significance for a 6-year window, and the histogram (bottom right) shows most of the residuals have a near-zero value and line up reasonably with the chi-squared distribution (the orange curve).

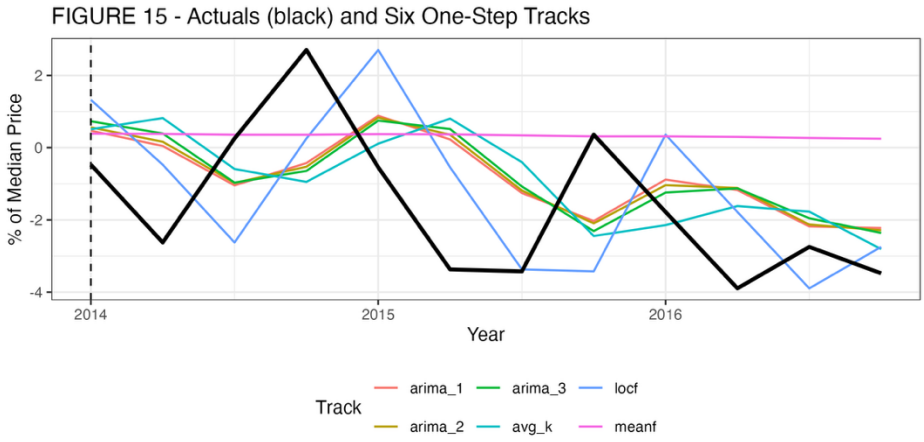


Section 7 – Forecasting

The models selected in previous sections now have to do a job. They will perform a rolling one-step-ahead forecast. As stated in Section 5, the series was divided into 3 sections: train, weights, and holdout. This section uses the three top-performing models trained on the first 132 observations to make predictions about weights sample of the series. Forecasts are made for time period *t+1* based on the values at time *t*. As the actual values are put into the model, it makes one new forecast for each next singular time period. This is equivalent to a weather reporter who only gives tomorrow's forecast.

To judge the success of these models there are three types of averages to compare them against. The last observation carried forward (locf) is the "average" of observation y_t used as a prediction for y_{t+1}. This rests on the assumption that the value of a time series today is likely to be close to its value tomorrow. The next metric (average k) takes the previous *k* values and averages them together. In the case of this series, k is set to 3, since the PACF retained up to 3 significant lags. This implies a somewhat persistent memory in the series, so a rolling window of 3 lags should capture the most recent memory of the data. Finally, the expanding mean (meanf) is simply the average of all observations up to the point of the forecast. If the ARIMA models do not perform better than the computationally cheaper averaging family, then it would be difficult to justify their use.

Figure 15 shows a plot of the results. The black line is the actual value from the data. The pink, nearly horizontal line is the meanf, and while it is close on average over the span of the whole series, the plot demonstrates how limited its use is in forecasting. The blue line is the locf, which looks like a shadow of the actual values. Then the red, yellow, and green ARIMA models follow a tight path that almost appear as a smoothed version of the locf. The teal line is the k-average curve. Had *k* been set lower, this line would converge to the locf curve, but with k=3, it tracks closer to the ARIMA models which have 2 and 3 lags.



Section 8 – Forecast Evaluation

With six methods of forecasting producing 12 forecasts each, the final step is to evaluate them. The three primary methods (or loss functions) used for measuring the accuracy of the predictions are mean squared error (MSE), mean absolute error (MAE), and root mean squared error (RMSE). Each loss function defines accuracy based on what is lost by being wrong. MSE and RMSE punishes errors exponentially, MAE penalizes them linearly, and all three metrics treat errors in both directions equally (i.e. an overestimate counts against the model the same as an underestimate of the same absolute value).

Given the context of the question and the scale of the values, the loss function most appropriate is the MSE. The RMSE and MSE will produce identical rankings, but at this scale, the MSE will be slightly easier to differentiate. The reason for choosing MSE over MAE is that the metrics at play are derived to imply a distribution of the largest assets ordinary people own. An error of 3 percentage points has much smaller effects than 6 or 12. Put formally the MSE is:

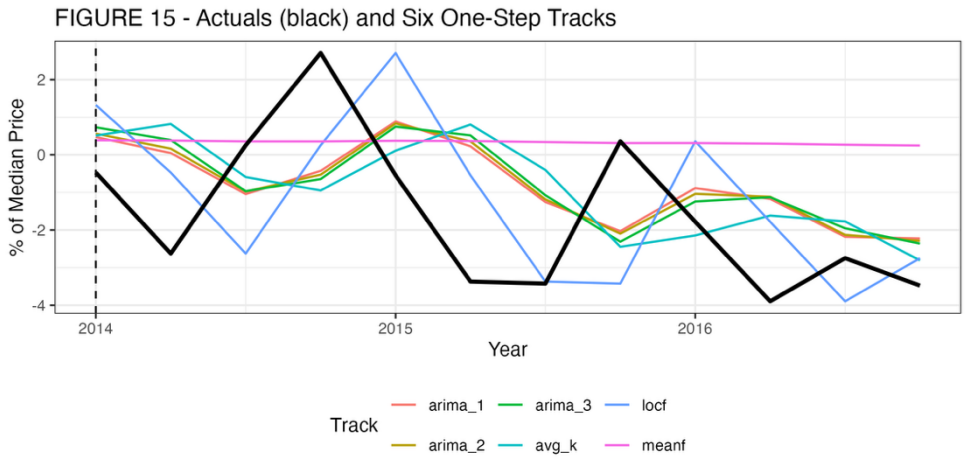
$$MSE = \frac{1}{N}\sum_{t=1}^{N}(y_t - \hat{y}_t)^2$$

In this case, N equals 12, y_t is the actual value at time *t* and $\hat{y}_t$ is the prediction at time *t*. The scores are given in Table 8 and reveal that the ARMA(1,1) is the most accurate model, but it could have been otherwise. The MSE values increase (get worse) from the first place ARIMA model to the third, according to the rankings from section 5. Of the naïve models, last observation carried forward did the best, average k was worse, and meanf had a significantly higher error rate that everything else. As mentioned before, RMSE ranks everything identically to MSE, but reports the square root. The MAE however, produces very different rankings.

TABLE 8 - Comparison of Models' Loss Functions

Model	MSE	RMSE	MAE
arima_1	4.58521	2.14131	0.78575
arima_2	4.81057	2.19330	0.79825
arima_3	5.20970	2.28248	0.81041
locf	5.49336	2.34379	0.40016
avg_k	5.70716	2.38897	0.71150
meanf	7.43507	2.72673	1.91816

The best model by far when using the MAE is the locf, and all of the ARIMA models perform even worse than the avg_k. Returning to Figure 15, the ARIMA bands are closer than the locf to actuals about half the time. When they are farther away, they tend to be only slightly farther. When the locf curve is farther, it tends to be much farther.



AI Disclosure

I used Claude Sonnet 5.5 Medium for help with this report. I asked for clarification on how the forecasting function worked, help with minor code adjustments, and confirmation that my forecasts were targeting the correct dates, and it helped clarify the formal definition of serial correlation. There was one function and code block that I copied from Claude, which was what I used to generate tables for the ADF and KPSS tests for each series, although I adjusted the aesthetics moderately. At one point, I was going to use Census data for Income inequality and had Claude clean that data significantly, but found more frequent data from Realtime Inequality and scrapped that.

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Bureau, US Census. "Historical Income Tables: Income Inequality." Census.Gov. Accessed October 6, 2026. https://www.census.gov/data/tables/time-series/demo/income-poverty/historical-income-inequality.html.

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Rodda, David Thomas. "Rich Man, Poor Renter: A Study of the Relationship between the Income Distribution and Low-Cost Rental Housing." Ph.D., Harvard University, 1994. https://www.proquest.com/docview/304106596/abstract/26C4326755564DC2PQ/1.

Appendix

Section 3 – Data

TABLE 1 - Data Sources and Summary

Series	*Source & Series ID*	*Frequency*	*Span*	*N*	*Units*	*SA*	*Transformation*	*Accessed*
Average Home Prices	ASPUS - US Census/HUD via FRED	Quarterly	1981Q1 - 2019Q4	156	2017 USD	Yes	Detrended	10/06/2026
Median Home Prices	MSPUS - US Census/HUD via FRED	Quarterly	1981Q1 - 2019Q4	156	2017 USD	Yes	Detrended	10/06/2026
Average - Median Home Prices	Constructed from ASPUS and MSPUS	Quarterly	1981Q1 - 2019Q4	156	Share of Median Price (%)	Yes	Detrended	10/06/2026
Top 10% Income Earners	Realtime Inequality	Quarterly	1981Q1 - 2019Q4	157	Share of Income Earned (%)	No	Diff. (1)	10/07/2026
Housing Inventory	CPS/HVS: Table 8 - US Census Bureau	Quarterly	1981Q1 - 2019Q4	157	Housing Units (millions)	Yes	Diff. (1)	10/21/2026

Section 4 – Stationarity

Average Home Price Figures and Tables

FIGURE 1 - Average Home Prices 1981 - 2019

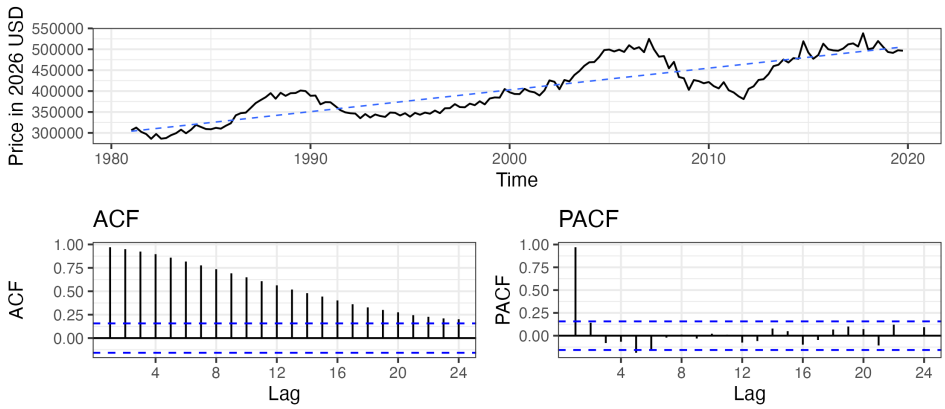


FIGURE 2 - Detreneded Avg. Home Prices

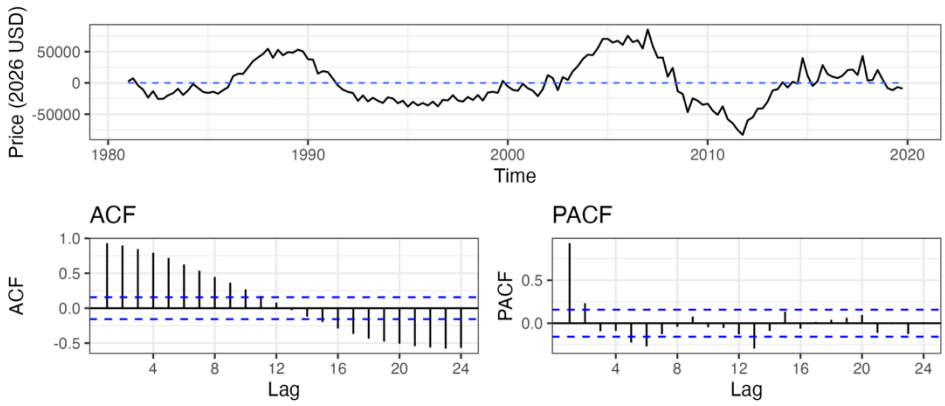


FIGURE 3 - Differenced Avg. Home Prices

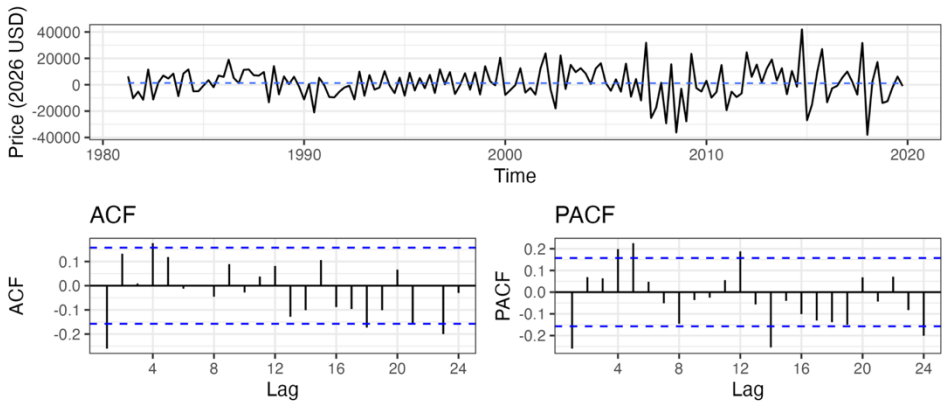


TABLE 2 - Unit-root and stationarity tests

Test	Null	Statistic	Critical values 1%	5%	10%	Decision at 5%
Average Home Prices						
ADF	Unit root	−0.905	−3.460	−2.880	−2.570	Fail to reject
KPSS	Stationary	2.476	0.739	0.463	0.347	Reject null
Detrended Avg. Home Prices						
ADF	Unit root	−1.794	−2.580	−1.950	−1.620	Fail to reject
KPSS	Stationary	0.108	0.739	0.463	0.347	Fail to reject
Differenced Avg. Home Prices						
ADF	Unit root	−9.067	−3.460	−2.880	−2.570	Reject null
KPSS	Stationary	0.059	0.739	0.463	0.347	Fail to reject

Median Home Price Figures and Tables

FIGURE 4 - Median Home Prices 1981 - 2019

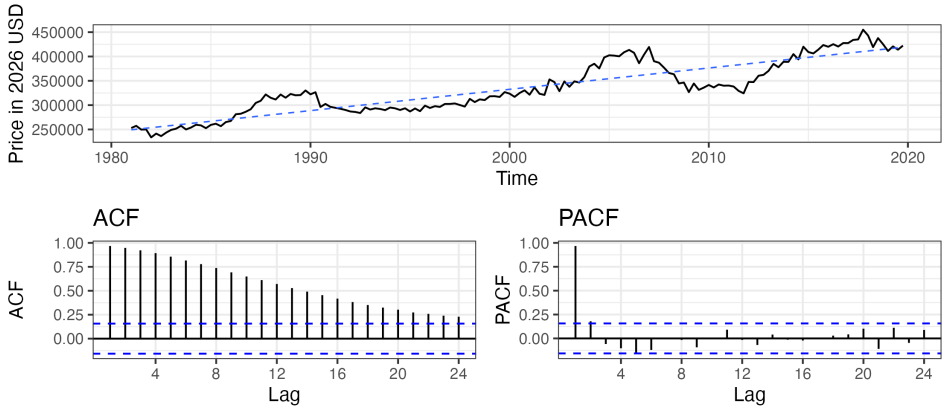


FIGURE 5 - Detrended Med. Home Prices

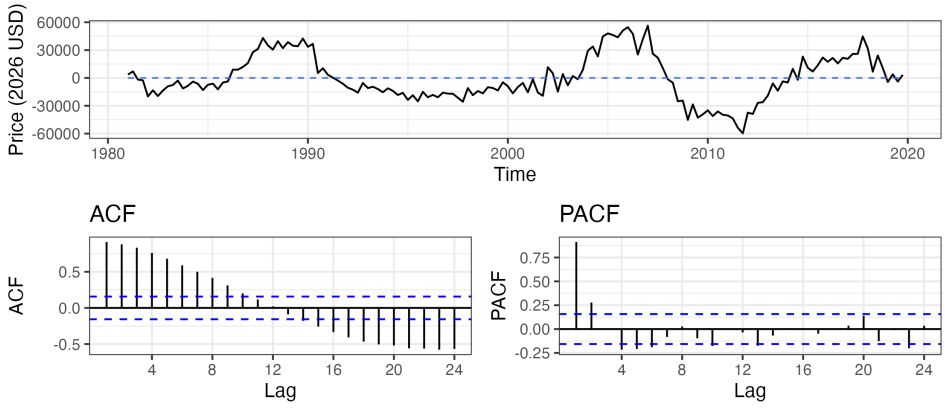


FIGURE 6 - Differenced Median Home Prices 1981 - 2019

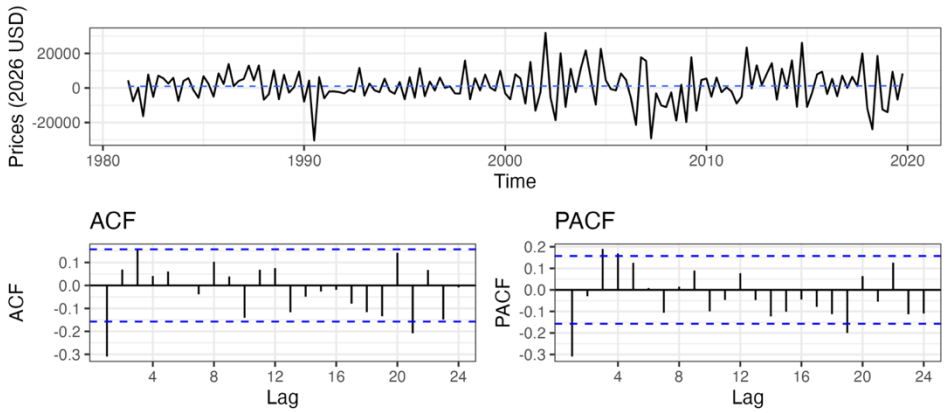


TABLE 3 - ADF and KPSS Test Results of Median Home Prices

			Critical values			
Test	Null	Statistic	1%	5%	10%	Decision at 5%
Median Home Prices						
ADF	Unit root	−0.712	−3.460	−2.880	−2.570	Fail to reject
KPSS	Stationary	2.597	0.739	0.463	0.347	Reject null
Detrended Med. Home Prices						
ADF	Unit root	−1.945	−2.580	−1.950	−1.620	Fail to reject
KPSS	Stationary	0.105	0.739	0.463	0.347	Fail to reject
Differenced Med. Home Prices						
ADF	Unit root	−10.202	−3.460	−2.880	−2.570	Reject null
KPSS	Stationary	0.051	0.739	0.463	0.347	Fail to reject

Home Price Skew Figures and Tables

FIGURE 7 - Skew of Home Price Distribution 1981 - 2019

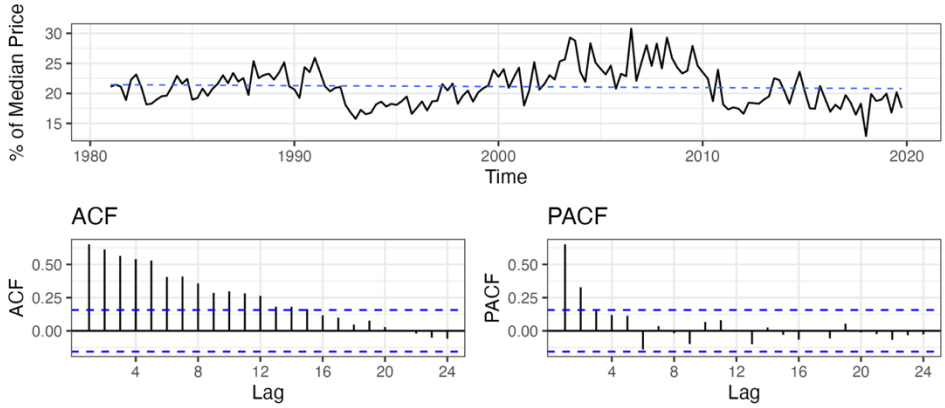


FIGURE 8 - Detrended Skew of Home Prices 1981-2019

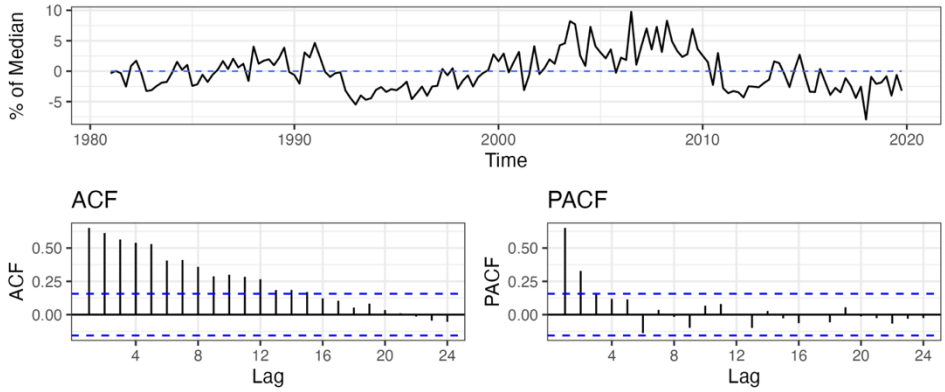


TABLE 4 - ADF and KPSS Test Results of Skew of Home Prices

			Critical values			
Test	Null	Statistic	1%	5%	10%	Decision at 5%
Skew of Home Prices						
ADF	Unit root	−3.569	−3.460	−2.880	−2.570	Reject null
KPSS	Stationary	0.274	0.739	0.463	0.347	Fail to reject
Detrended Skew of Home Prices						
ADF	Unit root	−3.603	−2.580	−1.950	−1.620	Reject null
KPSS	Stationary	0.280	0.739	0.463	0.347	Fail to reject

Income Inequality Figures and Tables

FIGIURE 9 - Income Share of the Top 10% of Income Earners

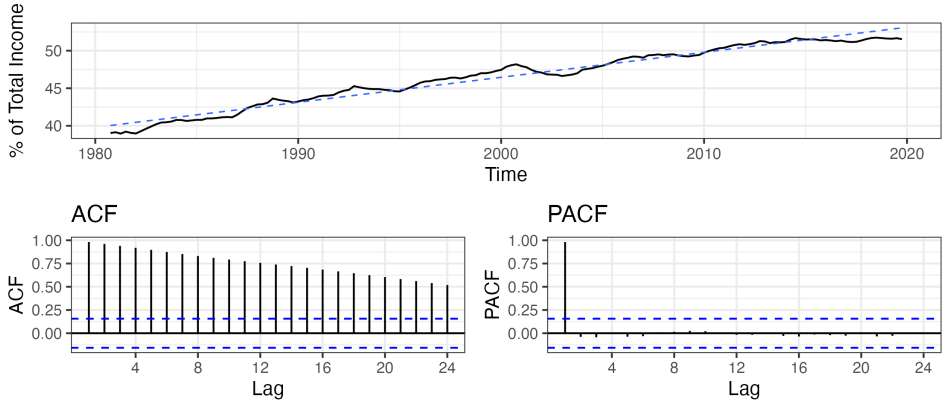


FIGURE 10 - Detrended Income Share of the Top 10% of Income Earners

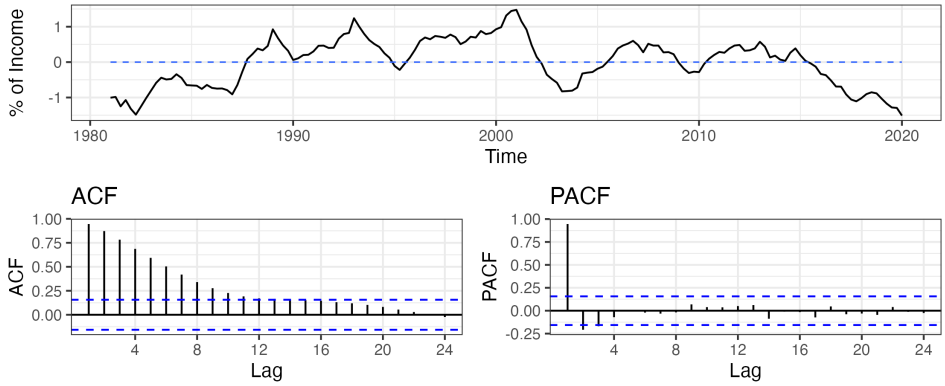


FIGURE 11 - First Differencing of Inequality

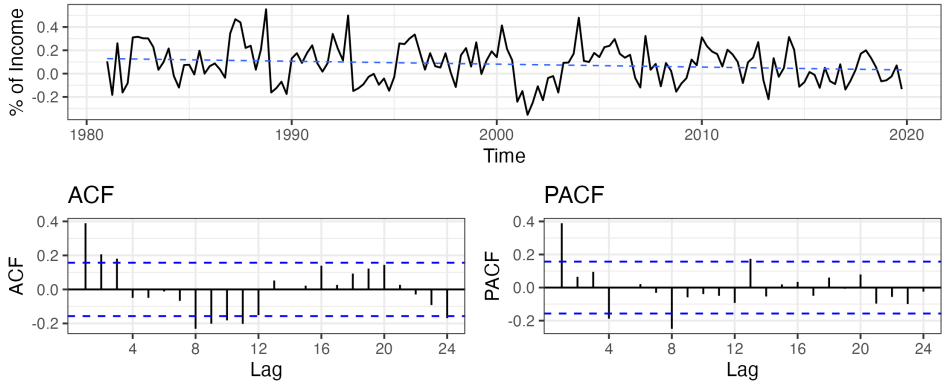


TABLE 5 - ADF and KPSS Test Results of Inequality

			Critical values			
Test	Null	Statistic	1%	5%	10%	Decision at 5%
Inequality						
ADF	Unit root	−1.784	−3.460	−2.880	−2.570	Fail to reject
KPSS	Stationary	3.144	0.739	0.463	0.347	Reject null
Detrended Inequality						
ADF	Unit root	−2.008	−2.580	−1.950	−1.620	Reject null
KPSS	Stationary	0.445	0.739	0.463	0.347	Fail to reject
Differenced Inequality						
ADF	Unit root	−6.358	−3.460	−2.880	−2.570	Reject null
KPSS	Stationary	0.239	0.739	0.463	0.347	Fail to reject

Housing Inventory Figures and Tables

FIGURE 12 - Housing Inventory 1981 - 2019

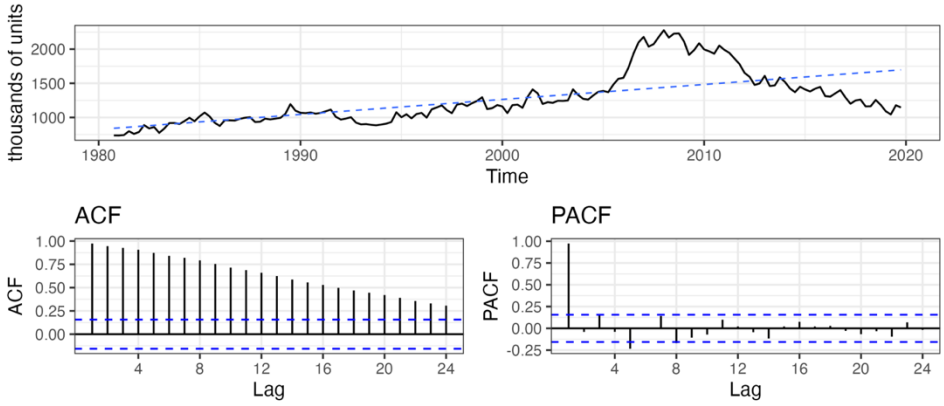


FIGURE 13 - Differenced Housing Inventory

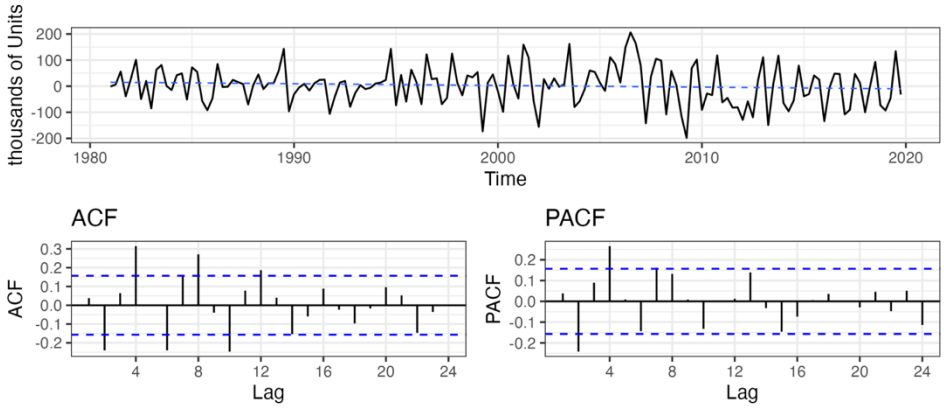


FIGURE 14 - Detrended Housing Inventory

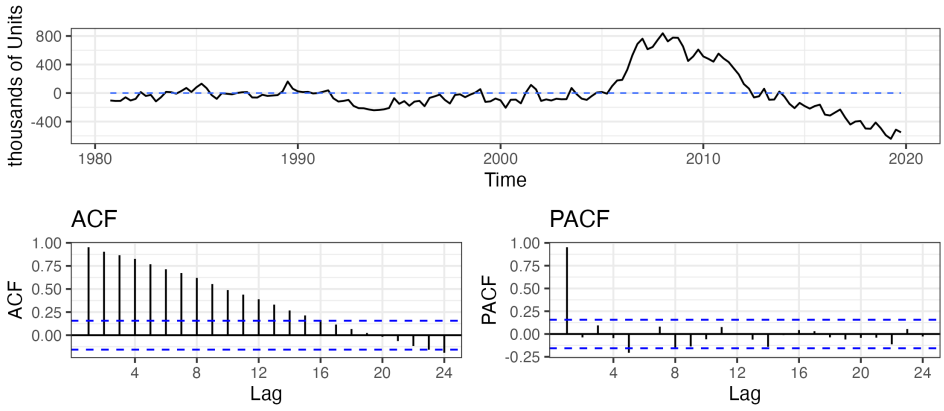


TABLE 6 - ADF and KPSS Test Results of Inventory

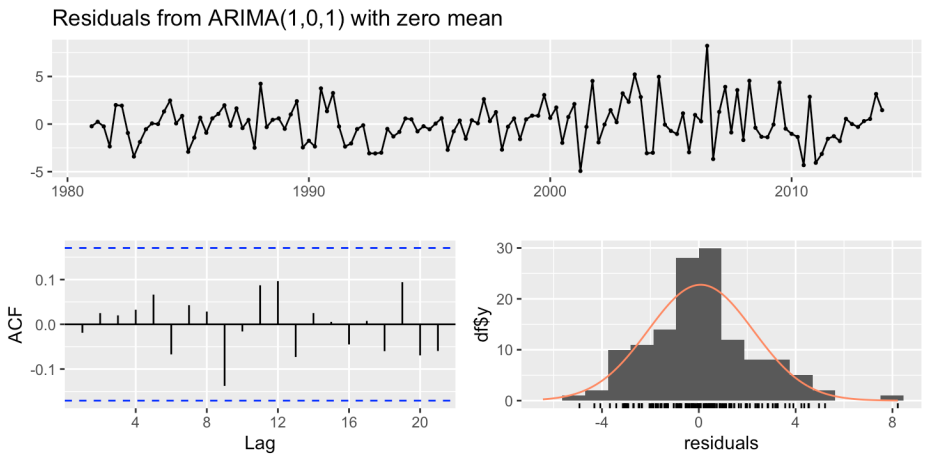
			Critical values			
Test	Null	Statistic	1%	5%	10%	Decision at 5%
Housing Inventory						
ADF	Unit root	−1.671	−3.460	−2.880	−2.570	Fail to reject
KPSS	Stationary	1.800	0.739	0.463	0.347	Reject null
Detrended Inventory						
ADF	Unit root	−1.226	−2.580	−1.950	−1.620	Fail to reject
KPSS	Stationary	0.302	0.739	0.463	0.347	Fail to reject
Differenced Inventory						
ADF	Unit root	−10.920	−3.460	−2.880	−2.570	Reject null
KPSS	Stationary	0.244	0.739	0.463	0.347	Fail to reject

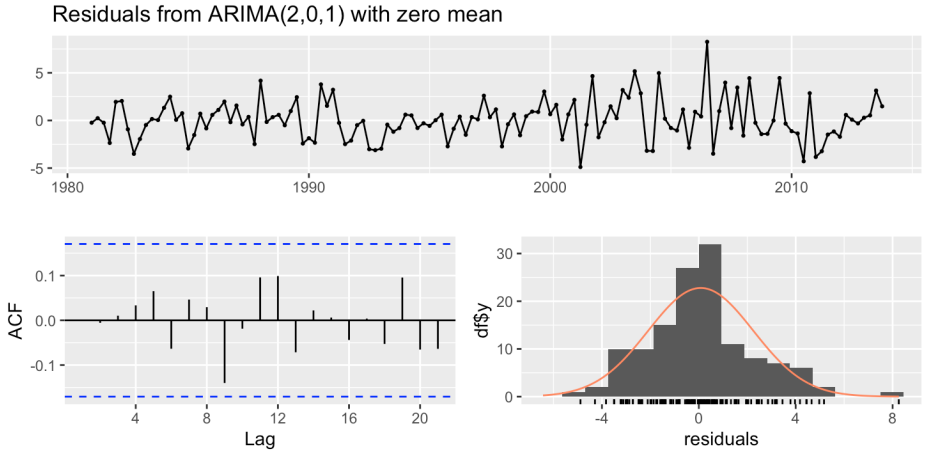
Section 5 – Model Selection

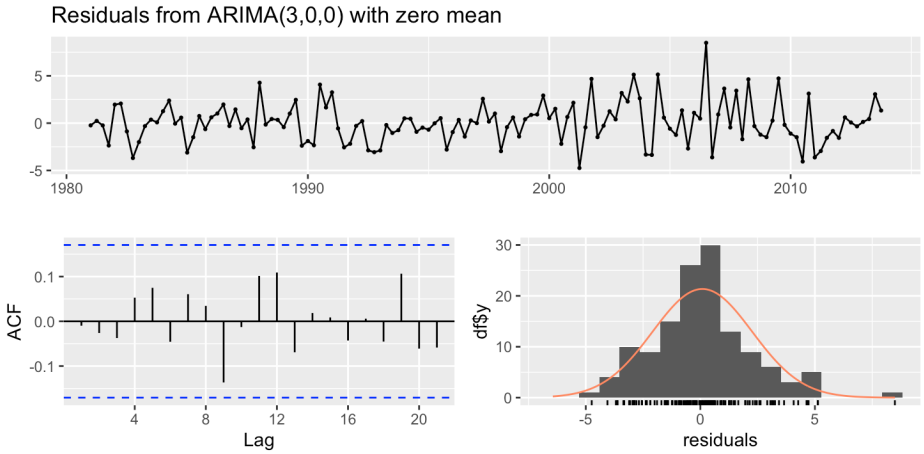
TABLE 7 - Comparison of Home Price Skew Models

Model	k	AIC	AICc	BIC
ARIMA(1,0,0)	2	601.8643	601.9573	607.6299
ARIMA(2,0,0)	3	588.9016	589.0891	597.5500
ARIMA(3,0,0)	4	587.8865	588.2014	599.4177
ARIMA(1,0,1)	3	585.4556	585.6431	594.1040
ARIMA(2,0,1)	4	587.2288	587.5438	598.7600

Section 6 – Serial Correlation

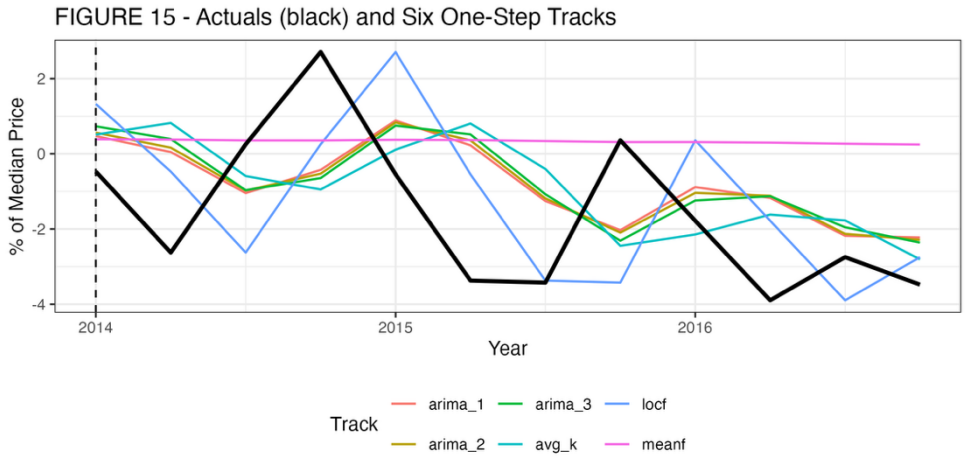





Residuals from ARIMA(3,0,0) with zero mean
5
0
-5
1980
1990
2000
2010
ACF
0.1
0.0
-0.1
4
8
12
16
20
Lag
df$y
30
20
10
0
-5
0
5
residuals

Section 7 - Forecasting

Forecasting Results



Section 8 – Forecast Evaluation

TABLE 8 - Comparison of Models' Loss Functions

Model	MSE	RMSE	MAE
arima_1	4.58521	2.14131	0.78575
arima_2	4.81057	2.19330	0.79825
arima_3	5.20970	2.28248	0.81041
locf	5.49336	2.34379	0.40016
avg_k	5.70716	2.38897	0.71150
meanf	7.43507	2.72673	1.91816